MIL-P-60620D(AR)
9 AUGUST 1985
SUPERSEDING
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9 November 1984

MILITARY SPECIFICATION

PROJECTILE ASSEMBLY FOR CARTRIDGE, SMOKE, WP, 81MM, M375A1

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements and provides quality assurance provisions for the fabrication of parts, assemblies and packing of projectile assembly for Cartridge, Smoke, WP, 81MM, M375A1.

2. APPLICABLE DOCUMENTS

2.1. Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER

10522511	- Projectile, 81MM, WP, M375A1 Assembly
7548319	- Box, Fiberboard Ammunition for Interplant Shipment of Projectiles, Empty

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI B 46.1 - Surface Texture

(Application for copies should be addressed to: American National Standards Institute (ANSI), 1430 Broadway, New York, NY 10018).

ASTM Method E-8 - Tension Test of Metallic Materials
ASTM Method E-10 - Brinell Hardness of Metallic Materials
ASTM Method B-117 - Salt Spray (Fog) Testing

(Application for copies should be address to the American Society for Testing and Materials (ASTM), 1916 Race Street, Philadephia, PA 19103).

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2.3 Order of precedence. In the event of conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. Materials, parts and assemblies shall comply with requirements specified on the applicable drawings, referenced specifications, and the following:

3.2 Parts. The parts shall comply with all requirements specified on drawing (dwg.) 10522511 as applicable, and with all requirements specified in applicable specifications and standards.

3.3 Metal defects.

3.3.1 Body. The body, including the cavity, shall be free from cracks, splits, bursts, solid shuts, pipe, porosity, inclusions, folds, seam and other metal defects.

3.3.1.1 Body cavity. The body cavity shall be free of scale, fins, burrs, draw marks, laminations, imbedded foreign matter, pits, and sharp edges.

3.3.2 Base adapter. The base adapter shall be free from cracks, splits, cold shuts, laminations, porosity or similar defects.

3.3.3 Visual standards (internal and external). Visual standards for evaluation of metal defects shall be established for each contract after the first months production or whenever requested by the procuring activity. However, the establishment of visual standards shall not waive the requirement for compliance with 3.3.1 and 3.3.2 .

3.4 Brazed joint (alternative body assmblies). The braze metal shall be visible for the full 360 degrees where the base adapter joins the body wall on the inside diameter. The braze metal shall also be visible for the 360 degrees around the joint between the rear of the body and the base adapter flange. Voids will be allowed on the exterior surface only of the joint, providing no single void is longer than 1/4 on an inch in length and the accumulated length of voids shall not exceed 1/2 inch of the outside periphery. Braze metal must be visible in all voids. Braze metal color shall not be cause for rejection.

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3.4.1 Hydrostatic test (alternative body assembly). The body assembly shall withstand stresses developed by the application of 5,000 pounds per square inch (psi) minimum, internal hydrostatic pressure for five seconds without leakage, rupture, or permanent deformation.

3.4.2 Air pressure test (alternative body assembly). After hydrostatic testing the body assembly shall withstand an internal air pressure of 150 psi minimum for 15 seconds minimum without evidence of leakage.

3.4.3 Joint quality (alternate body assembly). The brazed joint shall be achieved in a manner which will minimize entrapped slag, voids, unwetted or unbonded areas and other brazing defects. The aggregate area of these defects shall not exceed 25% of the area of the faying surfaces; and the maximum extent of a defect shall not exceed 50% of the overlap distance (width) of the joint.

3.5 Protective coating. Protective coating shall be in accordance with the applicable drawing and the specifications referenced thereon.

3.6 Ballistic performance (Proving Ground). The projectile body assembly shall reveal no evidence of failure when tested at excess and service pressure. Body assembly failure shall be:

a. Projectile body and base adapter deformation attributable to noncompliance with drawing and specification requirements.

b. Loss or breakup of any component, in the mortar tube or in flight (except obturator band).

3.7 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.8 Workmanship. The requirements for workmanship are as shown on the applicable drawings in the applicable specifications and the following:

3.8.1 Surface finish. The requirements for surface finish are as detailed on the applicable drawings. The cavity shall be free of burrs, scale, fins, deep seams, pits or other surface discontinuities.

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3.8.2 Threads. Threads shall be full and undamaged for the entire minimum length as required by the applicable drawings.

3.8.3 Burr. No part shall have a burr which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.8.4 Foreign matter. No part or assembly shall contain dirt, grease, chips, rust, corrosion, or other foreign matter. Particular attention shall be given to the cavity of the body assembly, to insure that no grit, scale, accumulated braze alloy, residual flux or other foreign matter remains therein after final cleaning.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First article inspection

b. Quality conformance inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the items in sample quantities as indicated.

Part Description	Drawing	Quantity
Metal parts assembly	10522511	55
Metal parts assembly (complete except paint)	10522511	5
Metal parts assembly (component parts unpainted, unassembled)	10522511	10

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-P-60620D(AR)

PARAGRAPH	TITLE: Projectile Assembly for Cartridge, Smoke, WP, 81MM, M375A1		SHEET 1 OF 1		DRAWING NUMBER: See below NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
I.	Metal Parts Assembly - Complete (See Note I) Including Paint (Dwg. 10522511)				
	a. Examination of defects	55	100%	3.2	4.4.2.4
	b. Proving Ground Tests	50	100%	3.6	4.5.6/4.4.4.1
	c. Salt Spray Tests	5	100%	3.5	4.5.5
II.	Metal Parts Assembly - Complete Except for Paint (Dwg. 10522511) (See Note 1)	5	100%	3.2	4.4.2.3
III.	Metal Parts Assembly, Component Parts, Unassembled (Dwg. D10522511) (See Note 1) a. Examination of defects	10	100%	3.2	4.4.2

NOTE I - The components and subassemblies shall be submitted to the examination for defects and tests in accordance with the paragraph reference column. All inspections and tests shall be performed 100%.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.1.1 Forging lots. Forging lots shall contain body forgings from not more than one heat of steel, that have been heat treated in the same heat-treatment equipment, at the same temperature, in one unchanged process. The forging lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.4.1.2 Cold shaping lots. Cold shaping lots shall contain cold shaped bodies, from not more than one heat of steel, that have been cold worked by the same process, utilizing the same tooling and equipment and stress relieved in the same furnace at the same temperature for the same length of time. The cold shaping lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.4.1.3 Miscellaneous lot. Small lots, such as bodies from abnormally small heats of steel or bodies which have become separated from previously delivered lots, etc., may, at the discretion of the manufacturer, be grouped into miscellaneous, forging, cold shaping or casting lots as the case may be. Such lots shall be identified by a miscellaneous lot code number.

4.4.1.4 Grouping. Forging, casting, or cold shaping lots shall remain grouped together for lot identity purposes, according to forging, casting or cold shaping lots, until such time as the results of the tension test have indicated that the lot has complied with the mechanical properties requirements of the applicable drawing.

4.4.1.5 Projectile lot. All forgings, casting or cold shaping lots that have been determined to have complied with the mechanical properties requirements of the applicable component drawing, may be re-grouped for assembly into projectile lots. A projectile lot shall consist of projectile assemblies produced by one manufacturer, under one contract, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision.

4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60620D(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Adapter, Base				10535863
					NEXT HIGHER ASSEMBLY 10522511
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of brazing surface		0.40%	3.2	Gage
102	Diameter of shoulder		0.40%	3.2	Gage
Minor					
201	Thickness of shoulder, min.		0.65%	3.2	Gage
202	Total length		0.65%	3.2	Gage
203	Angle of shoulder (if applicable)		0.65%	3.2	Gage
204	Radii missing or incorrect		0.65%	3.2	Visual (see 6.1)
205	Surface finish improper		0.65%	3.2	Visual (see 6.1)

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Body Alternative No. I				10535866
					NEXT HIGHER ASSEMBLY
					10522511
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Diameter of press fit surface		100%	3.2	Gage
2	Reverse taper of press fit surface		100%	3.2	Gage
3	Surface finish of press fit surface (indicated area)		100%	3.2	Gage/Visual (See 6.1)
Major	None defined				
Minor					
201	Total length		0.65%	3.2	Gage
202	Diameter at rear		0.65%	3.2	Gage
203	Radius at rear missing or incorrect		0.65%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.3	Projectile Subassembly (Prior to Paint and Insertion of Cap and VCI Paper)		1 OF 4	10535862/10535864
				NEXT HIGHER ASSEMBLY: 10522511

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Braze metal not visible for full 360 degrees at juncture of base adapter and body wall (alternative body only) (internal)		100%	3.4	Visual
2	Hydrostatic or air presure test stamps missing (alternative body assembly only)		100%	3.2	Visual
3	Metal defects		100%	3.2	Visual
4	Profile at intersection of radius with three degree angle and press fit surface for 360 degrees (as applicable)		100%	3.2	Visual
5	Braze metal voids in excess of requirements (external)		100%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Projectile Subassembly (Prior to Paint and Insertion of Cap and VCI Paper)	2 OF 4	10535862/10535864
			NEXT HIGHER ASSEMBLY: 10522511

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101	Weight		0.40%	3.2	Approved Scale
102	Runout of press fit surface with bourrelet		0.40%	3.2	Gage
103	Runout of band groove with bourrelet		0.40%	3.2	Gage
104	Runout of rear of body with bourrelet		0.40%	3.2	Gage
105	Runout of ogive with bourrelet		0.40%	3.2	Gage
106	Runout of pitch diameter with bourrelet including rear face.		0.40%	3.2	Gage
107	Perpendicularity of front face with bourrelet		0.40%	3.2	Gage
108	Diameter of band groove		0.40%	3.2	Gage
109	Diameter of band, min.		0.40%	3.2	Gage
110	Length from front face to rear of press fit surface		0.40%	3.2	Gage
111	Thickness of wall at bourrelet		0.40%	3.2	Gage
112	Thickness of wall between bourrelet and taper, min.		0.40%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Projectile Subassembly (Prior to Paint and Insertion of Cap and VCI Paper)	3 OF 4	10535862/10535864
			NEXT HIGHER ASSEMBLY: 10522511

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
113	Thickness of wall at front of taper		0.40%	3.2	Gage
114	Thickness of wall at rear of taper		0.40%	3.2	Gage
115	Thickness through base		0.40%	3.2	Gage
116	Thickness through base at juncture of thread boss and rear face, min.		0.40%	3.2	Gage
117	Width of band groove		0.40%	3.2	Gage
118	Surface finish of cavity improper		0.40%	3.2	Visual (see 6.1)
119	Foreign matter in cavity		0.40%	3.2	Visual
Minor					
201	Diameter of nose, outside		0.65%	3.2	Gage
202	Diameter at ogive basic length		0.65%	3.2	Gage
203	Diameter at rear bourrelet basic length		0.65%	3.2	Gage
204	Diameter at forward basic length on taper		0.65%	3.2	Gage
205	Diameter at rear basic length on taper (body only)		0.65%	3.2	Gage
206	Diameter of thread undercut		0.65%	3.2	Gage
207	Diameter at rear (body only)		0.65%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 4 OF 4		DRAWING NUMBER
4.4.2.3	Projectile Subassembly (Prior to Paint and Insertion of Cap and VCI Paper)				10535862/10535864
					NEXT HIGHER ASSEMBLY
					10522511
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
208	Length of thread boss		0.65%	3.2	Gage
209	Length from front face to rear of shoulder		0.65%	3.2	Gage
210	Length from front face to start of press fit diameter, max.		0.65%	3.2	Gage
211	Length from rear shoulder to band groove		0.65%	3.2	Visual
212	Total length, max.		0.65%	3.2	Gage
213	Surface finish improper (except as otherwise specified)		0.65%	3.8.1	Visual (see 6.1)
214	Overlap between body and base adapter (alternative design only)		0.65%	3.2	Visual
215	Radii or chamfer missing or incorrect		0.65%	3.2	Visual (see 6.1)
216	Burr		0.65%	3.8.3	Visual
217	Foreign matter (except as otherwise specified)		0.65%	3.8.4	Visual
218	Threads improper		0.65%	3.8.2	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Projectile Assembly (After Painting)	1 OF 1	10522511
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter of thread, min.		0.40%	3.2	Gage
102	Major diameter of thread, min.		0.40%	3.2	Gage
103	Diameter of bourrelet, max.		0.40%	3.2	Gage
104	Length of effective thread, min.		0.40%	3.2	Gage
105	Thread damaged		0.40%	3.8.2	Visual
106	Paint or phosphate on press fit surface		0.40%	3.2	Visual
107	Corrosion compound missing from press fit surface		0.40%	3.2	Visual
108	Foreign matter		0.40%	3.8.4	Visual
109	VCI crystals missing		0.40%	3.2	Visual
110	Protective cap missing		0.40%	3.2	Visual
Minor					
201	Pitch diameter of thread, max.		0.65%	3.2	Gage
202	Major diameter of thread, max.		0.65%	3.2	Gage
203	Corrosion preventative compound missing from thread		0.65%	3.2	Visual
204	Paint on thread		0.65%	3.2	Visual
205	Paint inadequate or damaged		0.65%	3.2	Visual
206	Marking incorrect, incomplete or illegible		0.65%	3.2	Visual

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Carton, Packing (Prior to Sealing)		7548319
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Support missing or damaged		0.65%	3.2	Visual
202	Separator missing or damaged		0.65%	3.2	Visual
203	Improper assembly		0.65%	3.2	Visual
204	Incorrect number in carton		0.65%	3.2	Visual

NOTES:

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Carton, Packing (After Sealing)				7548319
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Carton damaged, cut or gouged through all layers		0.65%	3.2	Visual
202	Carton inadequately sealed		0.65%	3.2	Visual
203	Contents loose		0.65%	3.2	Visual/Manual
204	Marking incorrect, incomplete or illegible		0.65%	3.2	Visual

NOTES:

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4.4.3 Inspection testing. The following tests shall be performed in accordance with the provision of 4.4.

4.4.3.1 Mechanical properties.

4.4.3.1.1 Hardness.

4.4.3.1.1.1 Forgings. In order to determine uniformity of heat treatment, each heat treated body will be subjected to a hardness test with a Brinell hardness tester using a five-second minimum dwell or by any other method approved by the procuring activity.

4.4.3.1.1.2 Cold shapings. Twenty-five cold shaped bodies from the first batch processed through the stress relief furnace for each cold shaping lot shall be subjected to a hardness test after stress relief. The hardness test shall be performed with a Brinell hardness tester, or by any other method approved by the procuring activity. Hardness values shall be recorded.

4.4.3.1.1.3 Miscellaneous. Each miscellaneous lot shall be subjected to the hardness tests listed in paragraphs 4.3.3.1.1.1 or 4.3.3.1.1.2, as applicable.

4.4.3.1.2 Tension test. Two bodies, one of the hardest and of the softest, from each forging, cold shaping or miscellaneous lot shall be subjected to this test to determine compliance with mechanical properties required by the applicable drawing. Failure of one or more tension test specimens to comply with the requirements shall be cause for rejection of the lot represented by the specimen.

4.4.3.2 Hydrostratic test (alternative body assembly). Prior to gaging, each body assembly shall be subjected to an internal hydrostatic pressure to determine compliance with 3.4.1. Failure of any assembly to comply shall be cause for rejection of the assembly. Body assemblies which comply shall be stamped with the letter "H" as indicated on the assembly drawing.

4.4.3.3 Air pressure test (alternative body assembly). Each body assembly that has passed the hydrostatic pressure test as indicated by the letter "H" stamped on the body, shall be subjected to a submerged air pressure test to determine compliance with 3.4.2. Failure of any assembly to comply as evidenced by the presence of air bubbles in the vicinity of the brazed joint or rising to the surface of the water shall be cause for rejection of the assembly. Body assemblies which comply shall immediately upon removal from the test fixture be stamped with the letter "A" as indicated on the assembly drawing.

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4.4.3.4 Joint quality (alternative body assembly). After air and hydrostatic testing but prior to the application of protective coating, 2 body assemblies from each lot shall be selected at random, sectioned and separated to determine compliance with paragraph 3.4.3. Failure of any specimen to comply shall be cause for rejection of the lot.

4.4.3.5 Protective coating. The sample size and acceptance criteria shall be in accordance with the assembly drawing and the requirements of the specifications listed thereon. Failure to comply with the requirements of the applicable specifications shall be cause for taking the corrective action recommended therein.

4.4.4 Ballistic performance. The following Proving Ground tests will be performed by the Government.

4.4.4.1 First article sample. The quantities listed in Table I shall be ballistically tested in the phases indicated.

TABLE I

PHASE	QUANTITY
Excess pressure	25
Service pressure	25

4.4.4.2 Production lot sample. The quantities listed in Table II shall be ballistically tested in the phases indicated.

TABLE II

PHASE	QUANTITY
Excess pressure	10 each lot
Service pressure	10 each lot

4.4.4.3 Excess pressure. Samples provided in accordance with Tables I and II will be subjected to this test. Failure of any projectile assembly to comply with 3.7 shall be cause for rejection of the first article sample or the production lot represented by the sample.

4.4.4.4 Service pressure. Samples provided in accordance with Tables I and II will be subjected to this test. Failure of any projectile assembly to comply with 3.7 shall be cause for rejection of the initial production sample or the production lot represented by the sample.

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4.4.5 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.4 herein.

4.5 Methods of inspection.

4.5.1 Mechanical properties.

4.5.1.1 Hardness test. Prior to finish machining, all bodies that will be subjected to the hardness test shall be prepared by grinding a flat on the cylindrical portion of the body where the obturating band groove will subsequently be machined. The test shall be performd using the equipment and procedures prescribed by ASTM Method E-10, except that the minimum dwell time shall be five seconds, when a Brinell Hardness Test is used.

4.5.1.2 Tension test. From the bodies selected in accordance with 4.4.3.1.2, two tension test specimens conforming to the largest obtainable standard "R" type specimen as prescribed by ASTM Method E-8 shall be taken longitudinally from each side of the body, 180° apart. Procedures and criteria for determining yield strength and elongation shall be as prescribed by ASTM Method E-8.

4.5.2 Hydr static test (alternative body assembly). The body assembly shall be hydrostatically tested using equipment capable of performing the test accurately and consistently. In performing the test, the pressure will be applied gradually until the required minimum pressure is reached, after which the pressure will be held for a minimum of five seconds. The equipment shall be calibrated prior to the start of each day's operation and at intervals of not over two hours of continuous operation. It shall also be calibrated prior to re-starting the operation after a layoff of one hour or more.

4.5.3 Air pressure test (alternative body assembly). This test shall be made using equipment capable of performing the test accurately and consistently. Body assemblies shall be mounted in an approved fixture in such a manner that the body assembly shall be completely submerged in water; the required air pressure shall be applied and held for a minimum of 15 seconds. The water in the tank shall be permitted to settle after immersion of the assemblies prior to the start of timing. Observation for evidence of leakage shall be made for the entire time the assembly is required to be pressurized. The equipment shall be calibrated at the start of each day's operation and at intervals of four hours and after a stoppage of three hours or more.

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4.5.4 Joint quality (alternative body assembly). The sample selected in accordance with paragraph 4.4.3.4 shall be sectioned and separated as follows: (a) the rear portion of the projectile assembly, one inch minimum forward of the base adapter shall be removed in such a manner that no portion of the brazed joint will be damaged, (b) the rear portion of the body containing the base adapter shall then be cut vertically through the center of the base adapter, (c) each portion of the body shall be peeled from the base adapter section. After peeling, the brazed area of the two halves of the base adapter and the mating pieces of the body shall be visually examined to determine compliance with 3.4.3.

4.5.5 Protective coating. The procedure for testing the cleaning of components and the application of protective coating shall be in accordance with the requirements of the applicable drawing and the specifications listed thereon. The salt spray test shall be performed utilizing equipment prescribed by ASTM Method B-117.

4.5.6 Proving ground functional tests.

4.5.6.1 Excess pressure. The samples supplied in accordance with 4.4.4, inert loaded to the service weight and dummy fuzed, will be assembled into complete rounds and fired for recovery in a mortar for which the round is standard, the tube of which will have a minimum of 25 percent life remaining based on current mortar tube condemnation criteria. A propellant charge adjusted to produce 109 to 117 percent of the upper pressure limit of the propellant used with the application projectile. Photographic equipment will be set up near the muzzle to record the flight of the projectile to determine compliance with 3.6. After three consecutive production lots have been tested successfully the excess pressure phase may be eliminated, however implementation of the excess pressure phase can be instituted whenever directed by procuring contracting officer."

4.5.6.2 Service pressure. The samples supplied in accordance with 4.4.4, inert loaded and fuzed will be assembled into complete rounds utilizing the service charge and fired in a mortar for which the round is standard for comparison with an equal number of rounds from a reference lot. The sample round and reference rounds will be fired alternately. Horizontal and vertical distance will be recorded for each round fired for determination of horizontal and vertical probable error (PE) in mils to assess accuracy. Data obtained will be recorded for information only and forwarded to the cognizant technical agency.

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5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with the requirements of Level C in MIL-STD-1169 and drawing 7548319.

6. NOTES

6.1 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability. The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis for surface roughness determination.

6.1.1 Visual standards. Visual standards to be established at contractors plant in accordance with 3.3.3.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of test data. The following addresses are to be added to the Ballistic Test Request (BTR) for distribution of firing test reports to the technical activity: Commander, US Army Armament, Research and Development Center, ATTN: AMSMC-QAT-B and SMCAR-LCU-SI, Dover, NJ 07801-5001.

6.4 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs, as required to: Commander, US Army Armament, Research and Development Center, ATTN: AMSMC-QAT-I, Dover, NJ 07801-5001.

6.5 Drawings. Drawings listed in Section 2 of this specification under the headings US Army Armament Research and Development Center may also include drawings prepared by, and identified by as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal or ARRADCOM drawings. Technical data originally prepared by these activities is now under the cognizance of AMCCOM.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
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(Project 1315-A666)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-60620D	PROJECTILE ASSEMBLY FOR CARTRIDGE, SMOKE, HP, 81MM, M375A1

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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